ADITYA PRAKASH

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EDUCATION

Emory University **May 2025**
Master's, Computer Science *GPA: 3.8*
- Received a merit scholarship to attend the program.

PROFESSIONAL EXPERIENCE

Appteon Inc. **Reston, VA, USA**
AI Software Engineer *Jul 2025 - Present*
- Engineered an LLM-powered medical assistant with a custom RAG pipeline, automating patient data review with diagnosis suggestions and medication conflict detection across 600+ cases weekly.
- Fine-tuned AI models on AWS SageMaker using Python and PyTorch, integrating them into a conversational pipeline with WebSockets and AWS Transcribe to process doctor–patient dialogues, and generate clinical notes.
- Developed and deployed a containerized full-stack platform (React, Next.js, Bun, Docker, AWS EC2, ECR, ECS) with CI/CD pipelines across three environments, documenting end-to-end workflows to ensure stable and efficient delivery of the medical assistant system.
- Built TypeScript REST APIs for clinical workflows. Implemented OAuth2/OIDC user authentication via Amazon Cognito (JWT/JWKS) and policy-driven RBAC middleware to enforce least-privilege, auditable access.
- Collaborated closely with medical professionals to ensure outputs were interpretable and clinically relevant. Led training sessions and created non-technical walkthroughs to help clinician teams adopt new workflows seamlessly.

Emory University **Atlanta, GA, USA**
AI Software Engineer *May 2024 - May 2025*
- Fine-tuned and deployed LLaMA 70B on a Flask server to power an AI-assisted screening system for evaluating applications. Reduced review time by 66% (from ~30 min. to ~10 min. per applicant) and saved ~$50,000 annually.
- Developed a faculty management system to track and approve leaves, performance reviews, awards, and tenure progression, streamlining administrative processes for 5,000+ faculty members, using React.js, Vite, Remix, and Next.js.
- Designed and implemented automatic grading software in React.js and Node.js for Python and Java assignments, automating grading for 300+ students and reducing grading time by 50% for faculty and TAs.

Watto AI (Y Combinator S23) **Remote**
Software Development Intern *Jun 2023 - Aug 2023*
- Used React.js, Next.js, TypeScript, and CSS to develop end-to-end features for 5+ core modules of an AI-assisted document editor, including document hub, forms, and outline editor, improving workflow efficiency for 200+ users.
- Implemented a real-time outline feature, allowing users to instantly visualize and restructure their document hierarchy while receiving AI-driven suggestions, reducing revisions and API calls by 58%.

Rakuten **Indore, Madhya Pradesh, India**
Software Development Intern *Jan 2023 - May 2023*
- Used Python and C++ to develop 50+ APIs for an AI-powered network capacity manager, automating key operations and improving system efficiency.
- Developed Shell and Python scripts to automate codebase setup, dummy data generation, and database querying, achieving a 50% reduction in setup and execution time and significantly streamlining testing pipelines on new systems.

PROJECTS

Canvas MCP Assistant - *Link to Project* **Atlanta, GA, USA**
- Improved accessibility of AI-driven tools for students by developing a Python tool that creates an MCP-compliant server integrating AI assistants with Canvas LMS, structuring course data into knowledge graphs and storing it in SQLite.
- Built an AI-powered lecture material retrieval system, enabling instant access to summarized content and lecture insights, improving revision speed by 2× for the average student and ensuring comprehensive study resources for students.

SKILLS

Programming Languages: Python, JavaScript, TypeScript, C, C++, Java
Databases: FAISS, Milvus, S3 Vector, PostgreSQL, MySQL, MongoDB, Redis
Agentic AI & LLMOps: vLLM, LangChain, Transformers, Tensorflow, PyTorch, Scikit-learn, NumPy, Natural Language Processing (NLP), Pandas, SciPy, Retrieval Augmented Generation (RAG), LoRA, PEFT, Model Context Protocol (MCP)
AI Safety, Evaluation, & Monitoring: Differential Privacy, Adversarial Robustness, Prompt Injection Mitigation, LangSmith
Full-Stack: Next.js, React.js, Vite, Express.js, Bun, Node.js, Flask, FastAPI, Redux.js, REST APIs, GraphQL, JWT, OAuth 2
Additional Tools & Platforms: Git, Bash, Linux/Unix, Docker, Kubernetes, AWS, EC2, S3, IAM, RabbitMQ, Socket.io